Section B

Answer **one** question from this section.

4 **(a)** Glycolic acid, $CH_2(OH)COOH$, is an α-hydroxy acid used in some skincare products.

Sodium glycolate can be prepared by adding disodium hydrogen phosphate to a solution of glycolic acid in a cosmetic formulation. The reaction establishes the following equilibrium in water:

equilibrium 1 $\quad CH_2(OH)COOH + HPO_4^{2-} \rightleftharpoons CH_2(OH)COO^- + H_2PO_4^-$

The K_a values for $CH_2(OH)COOH$ and $H_2PO_4^-$ are given in Table 4.1.

Table 4.1

acid	K_a
$CH_2(OH)COOH$	1.48×10^{-4}
$H_2PO_4^-$	6.20×10^{-8}

(i) Write down the IUPAC name for glycolic acid. [1]

(ii) Identify the two different conjugate acid-base pairs in equilibrium 1. [1]

(iii) Use the K_a values in Table 4.1 to calculate the equilibrium constant, K_c, for equilibrium 1. [2]

(iv) In an experiment, a buffer solution of pH 4.00 is prepared using 50.0 cm^3 of 0.0500 mol dm^{-3} glycolic acid and x cm^3 of 0.100 mol dm^{-3} NaOH.

$$CH_2(OH)COOH + NaOH \rightarrow CH_2(OH)COONa + H_2O$$

Assume that all NaOH reacts with glycolic acid, what is the volume, x cm^3, of 0.100 mol dm^{-3} NaOH required to make the buffer? [4]

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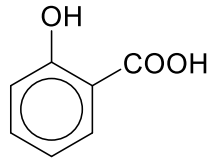
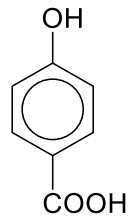
(b) Table 4.2 shows the pK_a values of two isomeric benzoic acids: salicylic acid and 4-hydroxybenzoic acid.

Table 4.2

acid	pK_{a1}	pK_{a2}
 salicylic acid	3.0	13.4
 4-hydroxybenzoic acid	4.1	9.7

Explain the following:

(i) pK_{a2} is larger than pK_{a1} of salicylic acid. [1]

(ii) pK_{a1} of salicylic acid is smaller than pK_{a1} of 4-hydroxybenzoic acid. [1]

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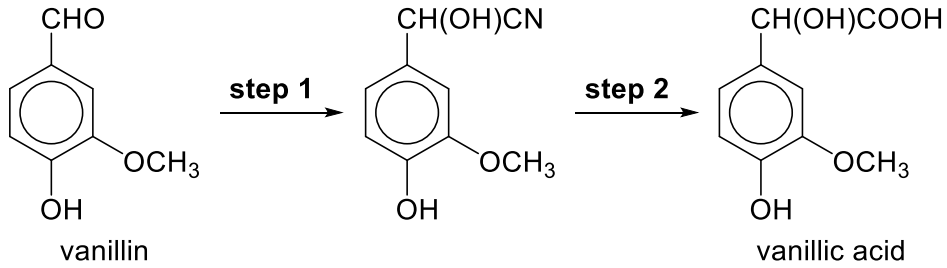
(c) Vanillin, a common flavor and fragrance ingredient, can be converted into vanillic acid by a two-step laboratory synthesis.



(i) State the reagent and conditions used in **step 1**. [1]

(ii) State the type of reaction that occurs in **step 1** and **2**. [2]

(iii) Draw the mechanism of the reaction in **step 1**. Show clearly all charges and the intermediate formed and use curly arrows to indicate the movement of electron pairs. [3]

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(d) Hexagonal boron nitride, h-BN, consist of planar sheets of alternate boron and nitrogen atoms, similar in arrangement to the carbon atoms in graphite. Its softness and sheen make powdered h-BN widely used in cosmetics.

(i) h-BN is often referred to as “white-graphite’. With reference to chemical bonding and structure, explain why both h-BN and graphite have a soft and slippery feel. [2]

(ii) Explain, with reference to electronegativity, why h-BN does not conduct electricity readily, whereas graphite does. [2]

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